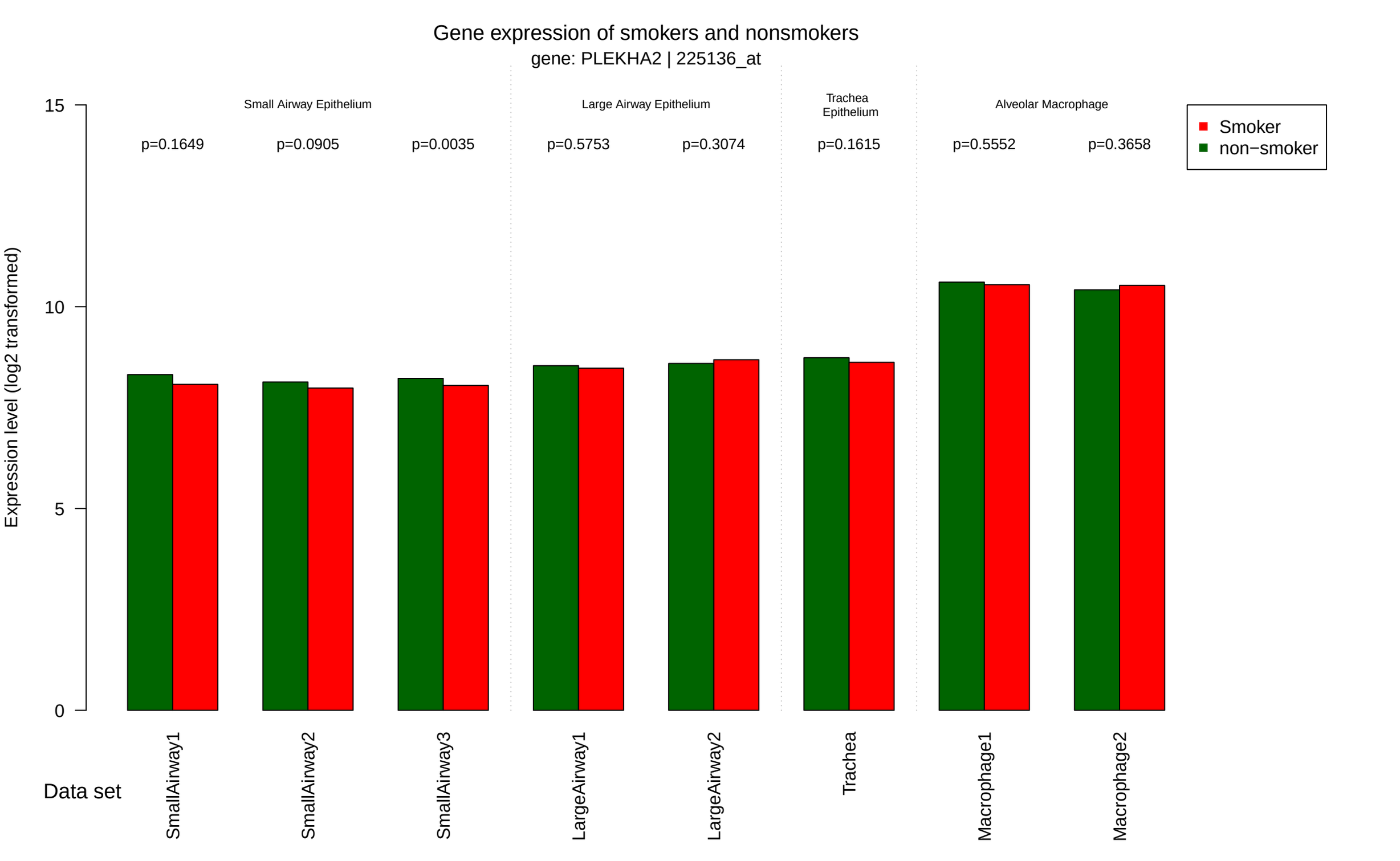
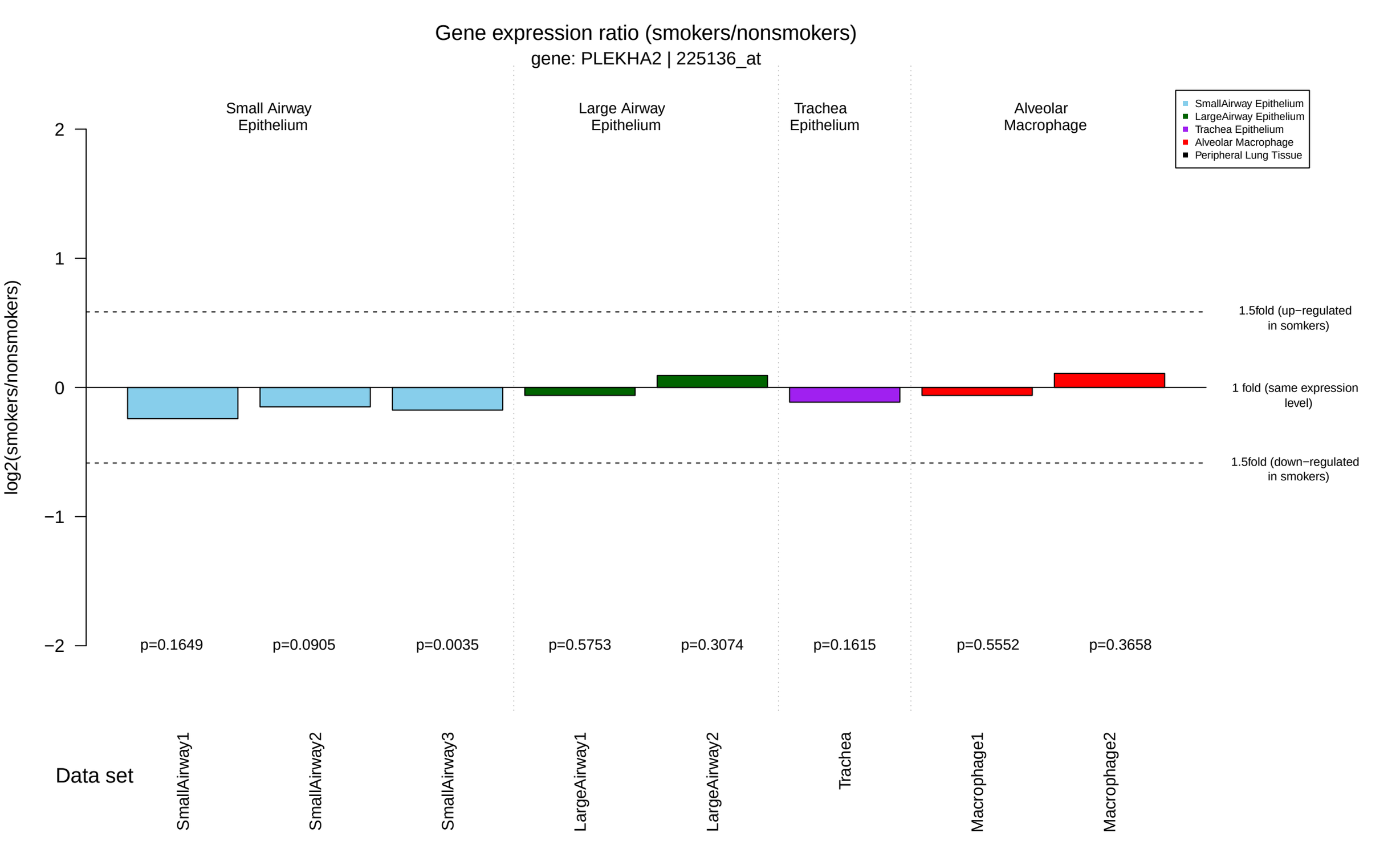
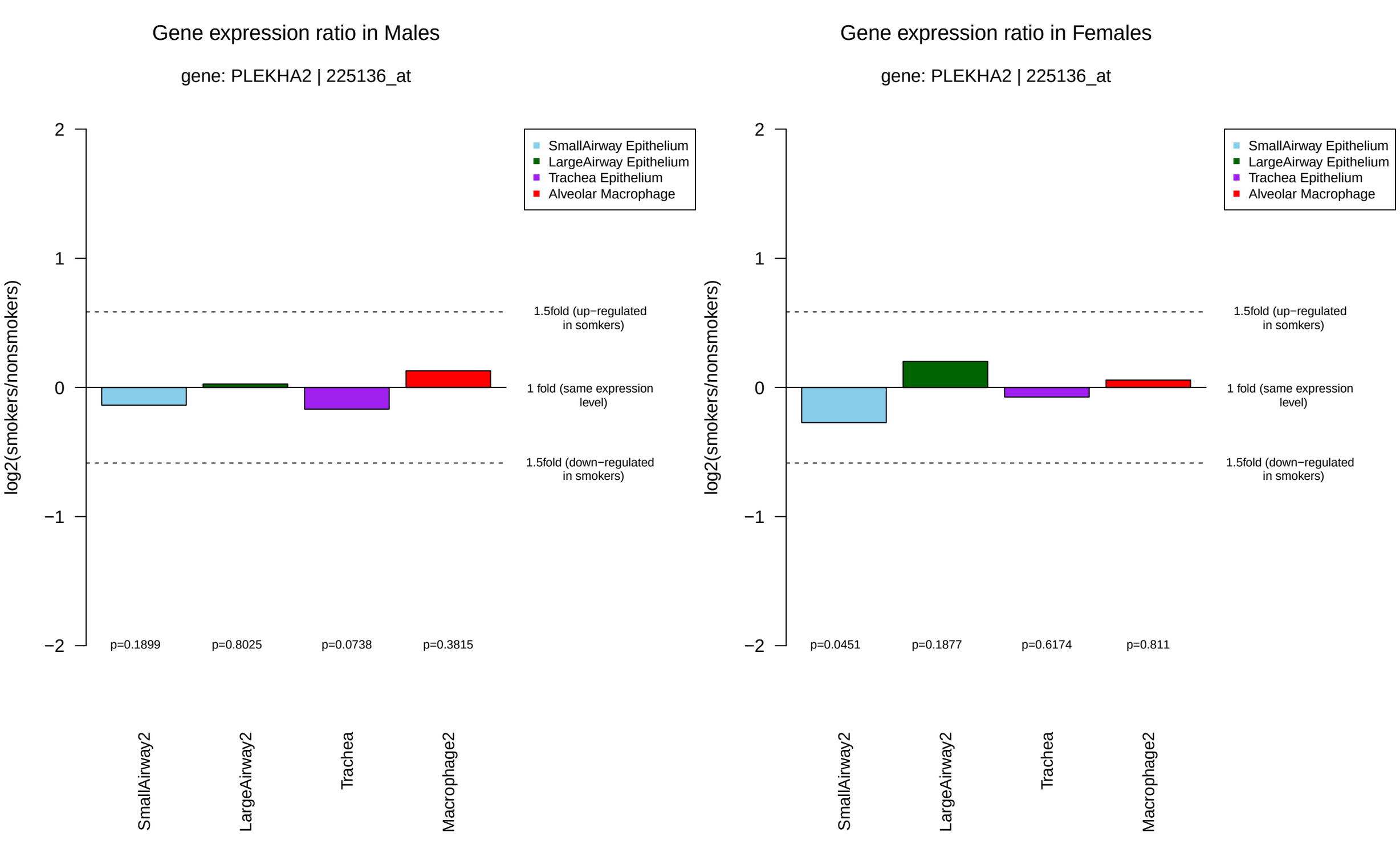
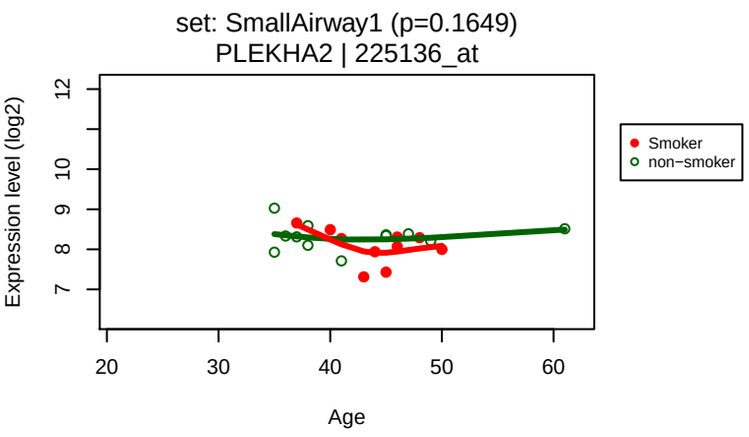
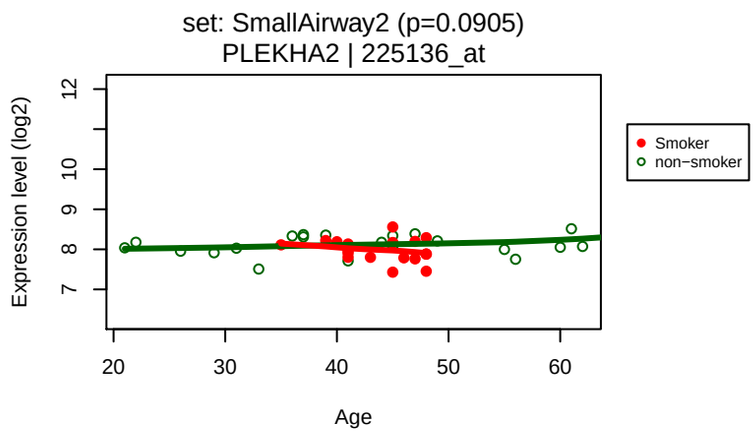
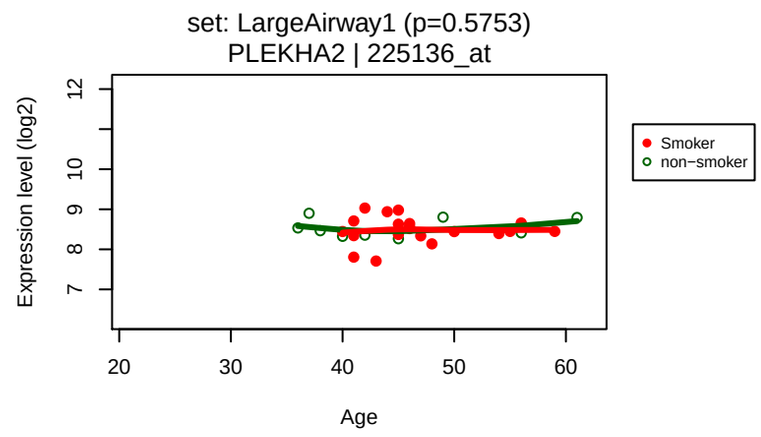

Gene expression of smokers and nonsmokers
gene: PLEKHA2 | 225136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1649
p=0.0905
p=0.0035
p=0.5753
p=0.3074
p=0.1615
p=0.5552
p=0.3658
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

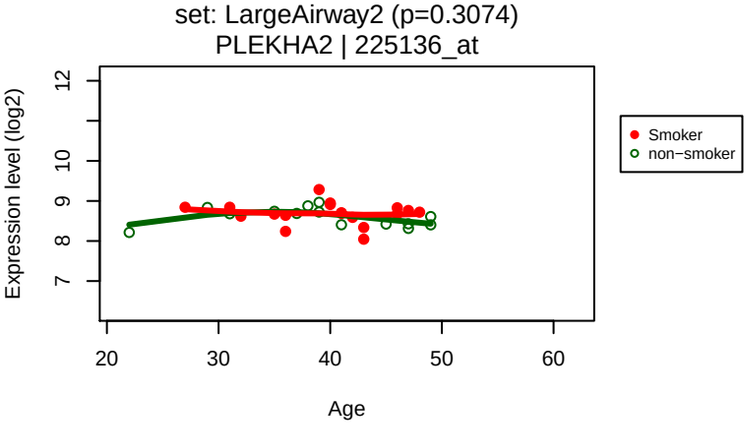

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHA2 | 225136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1649
p=0.0905
p=0.0035
p=0.5753
p=0.3074
p=0.1615
p=0.5552
p=0.3658
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

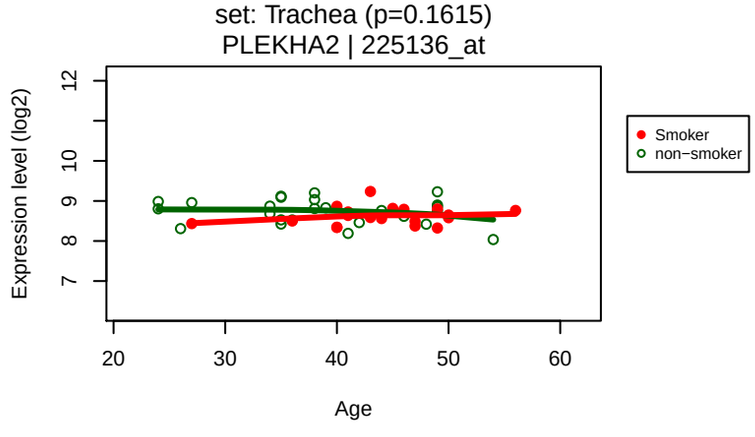

Gene expression ratio in Males
gene: PLEKHA2 | 225136_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1899
p=0.8025
p=0.0738
p=0.3815
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHA2 | 225136_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0451
p=0.1877
p=0.6174
p=0.811
SmallAirway2
LargeAirway2
Trachea
Macrophage2

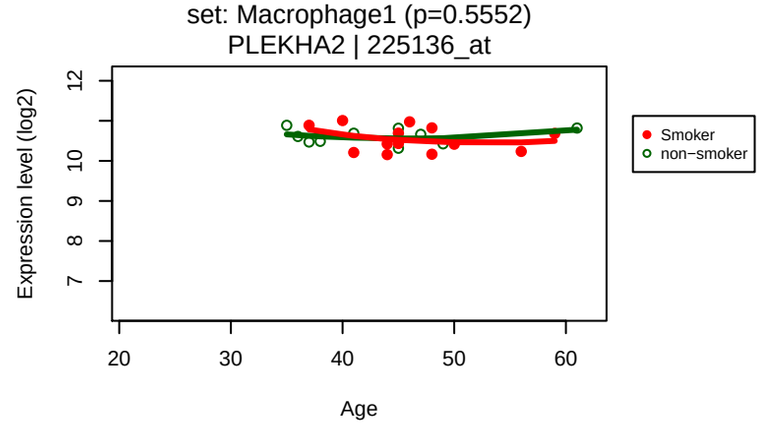

set: SmallAirway1 (p=0.1649)
PLEKHA2 | 225136_at
Expression level (log2)
Age
Smoker
non−smoker

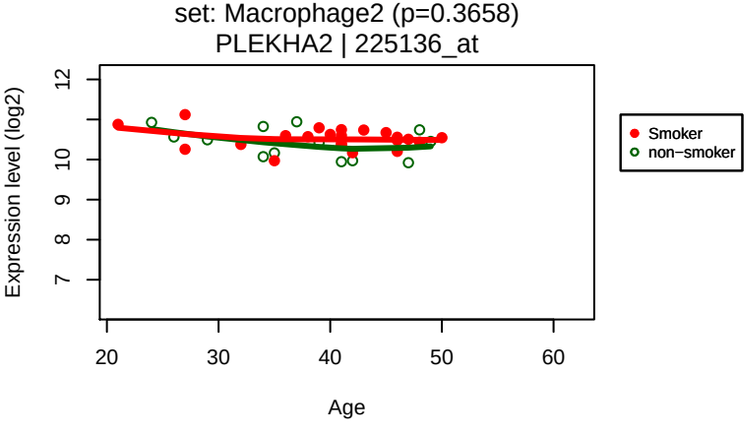

set: SmallAirway2 (p=0.0905)
PLEKHA2 | 225136_at
Expression level (log2)
Age
Smoker
non−smoker

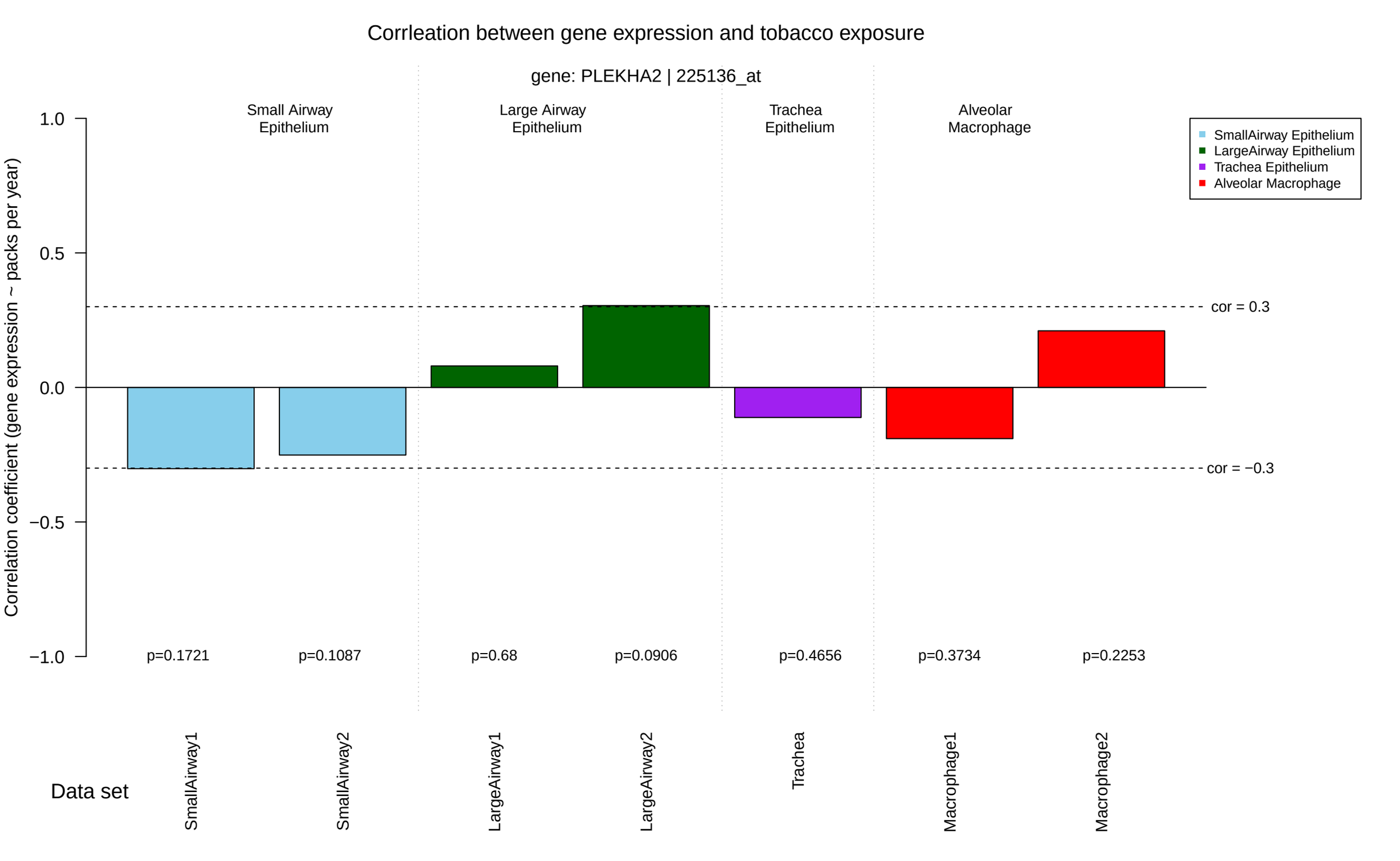

set: LargeAirway1 (p=0.5753)
PLEKHA2 | 225136_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3074)
PLEKHA2 | 225136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1615)
PLEKHA2 | 225136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5552)
PLEKHA2 | 225136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3658)
PLEKHA2 | 225136_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHA2 | 225136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1721
p=0.1087
p=0.68
p=0.0906
p=0.4656
p=0.3734
p=0.2253
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2